Stress Testing as a Financial Risk Measure

Jan Alsterlind

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Abstract

This memorandum examines stress testing as a financial risk measure, with specific reference to its role in the risk management of fixed-income portfolios at the Riksbank. While stress tests are often perceived as simple or deterministic tools for assessing losses stemming from changes in market prices, such as interest rates, the memorandum argues that stress tests are never assumption-free. Even when presented in a deterministic form, stress tests embed implicit views regarding the underlying pricing factors, their interdependencies across maturities, and their stochastic behavior.

By contrasting stress tests with distribution-based risk measures such as Value-at-Risk and Expected Shortfall, the memorandum clarifies the conceptual foundations of stress testing. Commonly used deterministic stress scenarios are interpreted as approximations of stochastic pricing relationships derived from no-arbitrage term structure models. Particular emphasis is placed on the role of pricing factors and the conditions under which stress scenarios can be regarded as economically meaningful.

The memorandum concludes that stress tests should primarily be interpreted as scenarios for changes in underlying pricing factors rather than as isolated price movements, and that transparency regarding these assumptions is a prerequisite for their appropriate use in the Riksbank's risk governance framework.

1. Introduction

A concept such as a *financial risk measure* can be defined with varying degrees of mathematical rigor. The purpose of this memorandum is not to provide a formal axiomatic definition of financial risk measures. Instead, a financial risk measure is here defined as a quantification of an unexpected loss that may occur in a portfolio over a given horizon.

Broadly speaking, financial risk measures can be divided into two main groups. The first group consists of measures that are explicitly based on a probability distribution of portfolio returns. These measures are widely used in financial risk analysis and their theoretical underpinnings are well documented in the literature. A comprehensive overview of their practical implementation is provided by Jorion (2007).

The second main group of risk measures consists of what are commonly referred to as *stress tests*. The term is not exclusive to financial economics but is also used in other disciplines. For example, Ellestad (2003) discusses stress testing in a medical context. Although stress tests are frequently used in financial institutions, their theoretical foundations are less systematically documented than those of distribution-based risk measures. As a result, the concept is applied in different ways, which may give rise to ambiguities regarding interpretation and use.

The purpose of this memorandum is therefore to clarify the assumptions on which stress tests as financial risk measures necessarily rest, even when the test itself appears simple or deterministic. The discussion provides a conceptual foundation for how stress tests should be interpreted and used within the Riksbank's risk management framework.

2. Two Main Types of Risk Measures

A financial risk measure is defined as a quantification of an unexpected decline in portfolio value. Let V_0 denote the value of a portfolio at an initial point in time and V_1 its value at a subsequent date. The change in value is defined as[1]

$$\Delta V = V_1 - V_0, \qquad V_1 < V_0.$$

Since the object of interest is the change in portfolio value, dV, risk measures can be classified into two broad categories. Both types are portfolio-dependent, meaning that they depend on how the portfolio is constructed. This dependence is denoted by a portfolio function $g(\cdot)$.

The first category includes distribution-based measures such as Value-at-Risk (VaR) and Expected Shortfall (ES), from which quantities such as Economic Capital may subsequently be derived [2]. These measures are based on the construction of a probability distribution over future portfolio returns, denoted by $g(y)$. The risk measure dV is then obtained directly from this distribution, for example as a particular percentile or tail expectation. Regardless of whether the return distribution is estimated parametrically, non-parametrically, or via simulation, these measures typically rely on realized historical outcomes of stochastic variables, namely portfolio returns.[3]

[1]The measure can also be defined in terms of a proportional change as $\Delta V/V = (V_1 - V_0)/V_0$.

[2]Strictly speaking, Economic Capital (EC) is not a standalone risk measure on the same conceptual footing as Value-at-Risk (VaR) or Expected Shortfall (ES). Rather, EC is typically defined as the capital requirement implied by a chosen risk measure, for example

$$EC = VaR_\alpha - \mathbb{E}[V].$$

For practical applications, the expected value term $\mathbb{E}[V]$ is often close to zero. This is the case for many of the Riksbank's risk exposures, implying that EC and VaR are numerically very similar.

[3]The method is based on the assumption that it is possible to construct a distribution for future returns. Such a distribution may either have an analytical form or be obtained through simulation;

The second category consists of stress tests. In this case, the focus is not on constructing a return distribution but on specifying pricing functions, $P(\cdot)$, that link underlying pricing factors (f) to portfolio returns, $y = dP(\cdot)$. The risk measure is defined as

$$dV = g(dP(df)),$$

that is, as the change in portfolio value induced by an assumed change in one or more underlying pricing factors. Stress tests therefore tend to focus on forward-looking scenarios and on the economic drivers of returns rather than on their historical distribution.

Despite this difference, stress testing as a risk analysis tool is not fully detached from historical outcomes. It is common for stress scenarios to be constructed with reference to a specific historical episode, such as the behavior of financial markets during the global financial crisis of 2008–2009. Similarly, distribution-based measures such as VaR may be estimated using return data from particularly turbulent periods. In this sense, stress tests and distribution-based risk measures may rely on similar information, even though they differ conceptually.

3. Stress Tests and Underlying Pricing Functions

A central component of stress testing is the explicit or implicit specification of pricing functions. Since the Riksbank's balance sheet primarily consists of fixed-income instruments, the discussion is henceforth restricted to a portfolio composed exclusively of bonds.

The pricing function for a zero-coupon bond with maturity $n = T - t$, face value F and continuously compounded yield is given by

$$B_n(y, t) = Fe^{-ny(t)}.$$

This pricing function depends on two arguments: the yield to maturity y and the remaining time to maturity $t < n$. A total differentiation of the bond price with respect to these arguments yields

$$dB = \frac{\partial B}{\partial y}dy + \frac{\partial B}{\partial t}dt + \frac{1}{2}\frac{\partial^2 B}{\partial y^2}dy^2 + \cdots$$

At this stage, the yield to maturity should be interpreted as an alternative quotation of the bond price. The expression above does not specify how yields on bonds of different maturities are related. From a purely technical standpoint, it is therefore possible to analyze the value change of, for example, a five-year bond independently of the value

however, the central parameters describing the distribution are fundamentally unknown. Consequently, the method relies on estimating these unknown parameters in some manner, and historical portfolio returns therefore almost always serve as the starting point for this estimation.

change of a four-year bond.

For stress testing to constitute a meaningful portfolio-level risk measure, however, it is necessary to account for the relationship between interest rates of different maturities. This requires an explicit or implicit term structure framework.

4. Value Changes and Underlying Pricing Factors

As previously noted, stress tests aim to quantify portfolio losses resulting from unexpected changes in one or more underlying pricing factors. For portfolios consisting of bonds with different maturities, a pricing framework is required that links yield changes across the term structure.

To illustrate this point, consider a simple setting in which the short risk-free interest rate i is the sole pricing factor, so that $f = i$. Together with time, the short rate determines the evolution of bond prices. Since pricing factors are assumed to be stochastic, the short rate is described by a stochastic differential equation:

$$di = \mu_t \, dt + \sigma_t \, dW_t,$$

where W_t denotes a Wiener process.

Let the price of a bond be represented by the function $B(i, t)$. By Itô's lemma, the change in bond value can be decomposed into a deterministic and a stochastic component:By Ito's lemma, the bond price yields a decomposition of the value change into a drift (deterministic) and a diffusion (stochastic) component:[4]

$$dB(i_t, t) = \underbrace{\left[\frac{\partial B}{\partial i}\mu_t + \frac{\partial B}{\partial t} + \frac{1}{2}\frac{\partial^2 B}{\partial i^2}\sigma_t^2\right] dt}_{\text{drift}} + \underbrace{\frac{\partial B}{\partial i}\sigma_t \, dW_t}_{\text{diffusion}}.$$

From a risk perspective, the deterministic component is not the primary object of interest. Instead, risk analysis focuses on the stochastic component, which links portfolio risk directly to the stochastic behavior of the underlying pricing factor.

In addition, a mechanism is required that relates the pricing factor to bond prices across maturities. This relationship can be represented generically as

$$B_n(t) = P\big(n, f(t)\big),$$

where $P(\cdot)$ denotes a generic pricing function relating maturity and underlying pricing factors to bond values. Term structure models derived from no-arbitrage principles provide economically consistent ways of specifying such relationships.[5]

[4]Under general assumptions, the equation corresponds to Itô's lemma; see Hull (2000), *Appendix 10A*.
[5]See Cox, Ingersoll, and Ross (1985) or Duffie and Kan (1996).

5. Deterministic Stress Tests as Approximations of Stochastic Relationships

In practical stress testing, the stochastic framework described above is rarely made explicit. Instead, stress scenarios are often defined directly in terms of changes in yields. A commonly used approximation expresses the proportional change in bond value as

$$\frac{dB}{B} \approx D\,dy, \qquad D = -\frac{1}{B}\frac{\partial B}{\partial y},$$

where D denotes duration.

This expression contains no explicit stochastic elements and can be interpreted as a special case of the more general pricing relationship. Under the assumptions that $\partial B/\partial y = \partial B/\partial i$ and that the stochastic shock $\sigma_d W_t$ is replaced by a yield change dy, the stochastic component of the pricing equation reduces to the duration-based approximation.

In many stress tests, it is implicitly assumed that yield changes are identical across maturities, so that all yields shift in parallel. This corresponds to the assumption that a single pricing factor affects the entire yield curve in the same way. Other stress scenarios, such as changes in the slope or curvature of the yield curve, similarly reflect changes in underlying pricing factors, even if these are not stated explicitly.

Although stress tests allow analysts to avoid explicit estimation of stochastic parameters based on historical data, the assumed yield changes cannot be entirely arbitrary. Stress scenarios are typically constrained by considerations of economic relevance and plausibility. At some level, therefore, deterministic stress tests must still be interpreted relative to the stochastic disturbances implied by the underlying pricing framework.

6. Stress Tests and Portfolio-Dependent Risk

The analysis above has implicitly assumed that the portfolio function $g(\cdot)$ is fixed. This assumption is common to most financial risk measures. In both risk analysis and investment decision-making, however, it is often of interest to assess how the outcome of a stress test would change under an alternative portfolio composition.

Let $\tilde{g}(\cdot)$ denote a counterfactual portfolio. The value change of this portfolio under a given stress scenario is

$$d\tilde{V} = \tilde{g}(P(df)),$$

which can be compared with the corresponding value change for the actual portfolio,

$$dV = g(P(df)).$$

Comparing stress outcomes across alternative portfolio compositions is an important component of both risk management and asset allocation analysis. For transparency and interpretability, differences between portfolios,

$$dg(\cdot) = \tilde{g}(\cdot) - g(\cdot),$$

should be expressed in terms of clearly defined portfolio characteristics such as duration, currency exposure, or asset allocation.

7. Conclusion

This memorandum has discussed stress testing as a financial risk measure in the context of the Riksbank's risk management framework. While stress tests may appear to be simple tools for quantifying losses resulting from changes in market prices, they are not free-standing assumptions about price movements.

Stress tests invariably embed assumptions regarding underlying pricing factors, their interdependencies, and their stochastic behavior, even when these assumptions are not made explicit. Stress tests should therefore be interpreted as scenarios for changes in underlying pricing factors rather than as isolated price movements.

For stress tests to be used effectively in risk governance, it should be possible to identify, assess, and communicate the assumptions on which they are based, including assumptions about factor structure, dependencies, and economic plausibility.

References

Jorion, P. (2007). *Value at Risk: The New Benchmark for Managing Financial Risk.* McGraw-Hill, New York.

Ellestad, M. H. (2003). *History of Stress Testing, Stress Testing Principles and Practice.* Oxford Academic.

Cox, J. C., Ingersoll, J. E., and Ross, S. A. (1985). A Theory of the Term Structure of Interest Rates. *Econometrica*, 53(2), 385–407.

Duffie, D. and Kan, R. (1996). A Yield-Factor Model of Interest Rates. *Mathematical Finance*, 6, 379–406.

Hull, J. C. (2000). *Options, Futures and Other Derivatives.* Prentice Hall.